A localized strict hierarchy for restricted arithmetic progressions

An expanded construction using the pronilfactor return-time theorem

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Abstract

For every $k \geq 1$ and every set $\mathcal{H} \subseteq \mathbb{N}$ whose complement is not piecewise syndetic, we construct a set $D_{k,\mathcal{H}} \subseteq \mathcal{H}$ that is k-large but not $(k+1)$-large. The positive conclusion holds even for colorings of a prescribed nilsystem return-time set. A single $(k+1)$-coloring blocks all progressions with difference in $D_{k,\mathcal{H}}$ and length $400 \cdot 2^k$, independently of $\mathcal{H}$. The sets can also be chosen to have zero upper Banach density and to be eventually divisible by every positive integer. Our main proof uses the non-piecewise-syndetic exceptional-set conclusion in Theorem A of Glasscock, Koutsogiannis, Le, Moreira, Richter and Robertson to accommodate the prescribed host. We give all details of the torus geometry, the moving-palette coloring, the finite extraction, and the mixed-radix assembly. The induction is uniform over the permitted sets $\mathcal{H}$; at each scale a fresh lower-color construction is chosen inside the appropriate dilated preimage of $\mathcal{H}$. A final comparison retains the almost one-to-one lifting argument from Qiu's Theorem B, explains the simplifications available for the unrestricted hierarchy, and shows how an independent syndeticity lemma also recovers the localized conclusion from Qiu.

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1 The statement and the plan

We write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. All arithmetic progressions in $\mathbb{N}$ have positive starting point and positive common difference. Their length is the number of terms, not the number of gaps.

Definition 1.1. A set $D \subseteq \mathbb{N}$ is *k-large* if, for every coloring $c : \mathbb{N} \to \{0, \dots, k-1\}$ and every $h \geq 2$, there are $b, d \in \mathbb{N}$ such that $d \in D$ and

$$c(b) = c(b+d) = \cdots = c(b+(h-1)d).$$

It is *large* if it is k-large for every finite k.

These notions were systematically studied by Brown, Graham and Landman [1]. We will separate every successive pair of color thresholds, with an additional restriction on where the allowed differences lie.

A subset of $\mathbb{Z}$ is *thick* if it contains intervals of arbitrarily large finite length. It is *piecewise syndetic* if finitely many of its translates have a thick union. When applying this definition to a subset of $\mathbb{N}$, we regard it as a subset of $\mathbb{Z}$. The *host* $\mathcal{H}$ below is the prescribed set in which we must construct our differences.

Theorem 1.2 (Localized hierarchy)**.** *Let $\mathcal{H} \subseteq \mathbb{N}$ and suppose that $\mathbb{N} \setminus \mathcal{H}$ is not piecewise syndetic. For every integer $k \geq 1$, there are a set $D_{k,\mathcal{H}} \subseteq \mathcal{H}$ and a coloring*

$$c_{k,\mathcal{H}} : \mathbb{N} \longrightarrow \mathbb{Z}/(k+1)\mathbb{Z}$$

with the following properties.

1. *$D_{k,\mathcal{H}}$ is k-large. More strongly, it has the nilsystem-window property $\mathcal{P}_k$ in Definition 2.1.*
2. *$c_{k,\mathcal{H}}$ has no monochromatic arithmetic progression of length*

$$H_k = 400 \cdot 2^k \tag{1.1}$$

   whose common difference belongs to $D_{k,\mathcal{H}}$.
3. *For each $m \geq 1$, the set $D_{k,\mathcal{H}} \setminus m\mathbb{N}$ is finite. In particular, $D_{k,\mathcal{H}}$ has zero upper Banach density.*

Thus $D_{k,\mathcal{H}}$ is not $(k+1)$-large. The forbidden length H_k is independent of the host $\mathcal{H}$.

The condition on $\mathcal{H}$ is equivalent to *thick syndeticity*: for each $\ell \geq 1$, the starting points of length-ℓ intervals contained in $\mathcal{H}$ have bounded gaps in $\mathbb{N}$. We prove this equivalence in Section 2.2. Consequently the theorem can be summarized as follows:

Every thickly syndetic set contains the entire strict color hierarchy.

Here "the entire hierarchy" means that for each k a suitable subset can be chosen; no nesting of those subsets is asserted. Taking $\mathcal{H} = \mathbb{N}$ recovers the unrestricted separation.

Example 1.3 (Prescribing excluded differences)**.** We may require all the constructed differences to avoid the squares, or any other fixed set of zero upper Banach density. Such an excluded set is not piecewise syndetic. For the squares, the number in any interval of length N is at most $\sqrt{N} + 1$, so their upper Banach density is zero. The implication from zero upper Banach density to non-piecewise-syndeticity is recalled in Section 9.4.

Remark 1.4 (Order of the quantifiers). The theorem says that for every host $\mathcal{H}$ a suitable set can be constructed *inside that host*. It does not say that one fixed D_k retains k-largeness after every deletion of a non-piecewise-syndetic set. Indeed our $D_{k,\mathcal{H}}$ itself has zero upper Banach density, so deleting that set would leave nothing. The distinction is important in the induction.

1.1 Why the proof has several parts

The positive and negative conclusions ask for opposite kinds of control. To prove k-largeness we must defeat every coloring with k colors. To prove failure of $(k+1)$-largeness we must exhibit one coloring with $k+1$ colors that works simultaneously for all permitted differences.

We first solve these two problems at a finite-dimensional stage. The dimension at that stage depends on the recurrence problem being tested, but the forbidden progression length does not depend on the dimension or the host. We then extract finite sets of differences and place them at widely separated integer scales. The separation of scales permits a single obstructing coloring to combine the local ones.

There are two distinct indices throughout the argument. The integer s counts colors in the induction. The integer j counts finite packets within one induction step. The localized version requires the old s-color recurrence set to depend on j, although its forbidden length remains fixed.

Part of the proof	Its role
Uniform induction hypothesis	Permits a fresh lower-color construction in each rescaled host.
Torus directions	Supply new differences while meeting nilsystem return conditions.
Moving palette	Adds one obstructing color without losing the old obstruction.
Return-time theorem	Preserves usable differences after excluding a non-piecewise-syndetic set.
Finite extraction	Handles a compact family of recurrence tests with finitely many differences.
Mixed-radix assembly	Combines packets without increasing the number of colors.

1.2 External inputs and the gain over the unrestricted proof

The principal input is Theorem A of Glasscock, Koutsogiannis, Le, Moreira, Richter and Robertson [3]. We use its linear specialization, stated precisely in Section 2.4. The nilsystem orbit-closure facts come from Leibman [5], and the countable parametrization uses rational-coordinate theory for nilpotent lattices [6, Chapter II]. The torus construction follows the idea in Alweiss's example [2]; its required finite-dimensional ingredients are proved here.

In the unrestricted proof, it suffices to lift one suitable recurrence configuration from a pronilfactor. For the localized theorem, a lifted difference must also avoid a prescribed excluded set. Theorem A gives exactly the additional control: among suitable factor return times, the failures of lifting form a set that is not piecewise syndetic. Such failures, together with the prescribed excluded set, cannot exhaust a piecewise syndetic supply of candidate differences.

The linear algebra, all coloring estimates, the set-theoretic facts about excluded sets, and the finite-packet and digit arguments are given below. The local coloring formulas and the final digit formula are explicit. The choices of lower-color sets and finite packets are existence choices; no computability claim or effective packet-size bound is made. Section 11 explains what can instead

be done with Qiu's Theorem B [4]. In particular, localization is not exclusive to Theorem A: Qiu together with the independent syndeticity lemma recorded as Lemma 3.6 of [3] also proves it. Theorem A gives the cleaner main argument and the sharper relative exceptional-set control described in Section 6.

2 The induction hypothesis and the dynamical input

2.1 Progressions inside a prescribed nilsystem window

A finite-dimensional nilsystem is a translation T on a compact nilmanifold $Y = G/\Gamma$, where G is a nilpotent Lie group and Γ is a discrete cocompact subgroup. We allow G and Y to be disconnected. For $y \in Y$ and an open neighborhood V of y, write

$$N_T(y, V) = \{n \in \mathbb{N} : T^n y \in V\}.$$

Definition 2.1. For $s \geq 1$, a set $D \subseteq \mathbb{N}$ has property $\mathcal{P}_s$ if, for every finite-dimensional nilsystem (Y, T), every $y \in Y$, every open neighborhood V of y, every coloring of $N_T(y, V)$ with at most s colors, and every $h \geq 2$, there is a monochromatic h-term progression in $N_T(y, V)$ with common difference in D.

Taking Y to be a one-point system shows that $\mathcal{P}_s$ implies s-largeness. The stronger formulation is needed because eliminating one color leaves the remaining colors available only at visits to an open nilfactor window. Ordinary s-largeness would not directly apply to this restricted set of times.

Lemma 2.2 (Dilation)**.** *If B has $\mathcal{P}_s$ and $a \in \mathbb{N}$, then aB has $\mathcal{P}_s$.*

Proof. Given a coloring of $N_T(y, V)$, color $n \in N_{T^a}(y, V)$ by the original color at an. Apply $\mathcal{P}_s$ of B in the nilsystem (Y, T^a). Multiplying the resulting progression of integer times by a gives the required progression for T, with difference in aB. □

The induction assertion we actually prove is uniform over hosts:

$$\begin{aligned}(\mathrm{I}_s): \quad &\text{for every } \mathcal{H} \subseteq \mathbb{N} \text{ with non-piecewise-syndetic complement,} \\ &\text{there is } B \subseteq \mathcal{H} \text{ with } \mathcal{P}_s \text{ and an } (s+1)\text{-color} \\ &\text{obstruction of length } H_s, \text{ where } H_s \text{ depends only on } s.\end{aligned} \tag{2.1}$$

The additional divisibility and density conclusions will follow from the assembly. The initial stage uses an empty old set and a constant old coloring; no property involving zero colors is required.

2.2 Piecewise syndeticity and the calculus of excluded sets

A set $A \subseteq \mathbb{Z}$ is *syndetic* if some integer $r \geq 1$ has the property that every interval of length r meets A. For subsets of $\mathbb{N}$ we say *syndetic in* $\mathbb{N}$ when only intervals in $\mathbb{N}$ are considered. A set syndetic in $\mathbb{N}$ is piecewise syndetic as a subset of $\mathbb{Z}$: finitely many translates cover a positive half-line, which is thick. It need not be syndetic in all of $\mathbb{Z}$.

Lemma 2.3 (Thickly syndetic complements)**.** *For $E \subseteq \mathbb{Z}$, the following are equivalent.*

1. *E is not piecewise syndetic.*

2. *For every $\ell \geq 1$, the set*

$$A_\ell = \{b \in \mathbb{Z} : \{b, b+1, \ldots, b+\ell-1\} \subseteq \mathbb{Z} \setminus E\}$$

is syndetic.

For $E \subseteq \mathbb{N}$, the corresponding one-sided condition says that $\mathbb{N} \setminus E$ is thickly syndetic in $\mathbb{N}$.

Proof. A set is not thick precisely when its complement is syndetic. Also E is piecewise syndetic precisely when, for some ℓ, the set

$$\bigcup_{i=0}^{\ell-1} (E - i)$$

is thick. Indeed every finite set of translation offsets is contained in a translate of a finite interval, and enlarging the set of offsets preserves thickness. The complement of the displayed union is exactly A_ℓ. This proves the equivalence. For subsets of $\mathbb{N}$, the finitely many boundary positions created by fixed translation offsets do not affect thickness or piecewise syndeticity. Equivalently, one can apply the same argument to positive intervals; the missing negative half-line creates no additional condition on their starting points. □

Lemma 2.4 (Excluded-set calculus)**.** *The following facts hold.*

1. *A subset of a non-piecewise-syndetic set is not piecewise syndetic. Finite unions of non-piecewise-syndetic sets are not piecewise syndetic.*
2. *If $P \subseteq \mathbb{Z}$ is piecewise syndetic and $E \subseteq \mathbb{Z}$ is not piecewise syndetic, then $P \setminus E$ is piecewise syndetic.*
3. *If $E \subseteq \mathbb{N}$ is not piecewise syndetic and $\lambda \geq 1$, then*

$$E^{(\lambda)} = \{d \in \mathbb{N} : \lambda d \in E\}$$

is not piecewise syndetic.

Proof. The subset assertion follows because piecewise syndeticity is upward closed. For finite unions, it is enough to prove that the intersection of two thickly syndetic sets is thickly syndetic. Let $A, B \subseteq \mathbb{Z}$ be thickly syndetic and fix a desired block length ℓ. Choose r so that every interval of length r contains the start of a length-ℓ block in B. Starts of length-$(r+\ell)$ blocks in A are syndetic. Within each such A-block, the choice of r supplies a length-ℓ block in B, hence a length-ℓ block in $A \cap B$. Its start lies within r places of the start of the A-block. Therefore the starts of the common length-ℓ blocks are syndetic. Lemma 2.3 now proves the finite-union assertion by taking complements.

For the second assertion, if $P \setminus E$ were not piecewise syndetic, then both sets on the right side of

$$P = (P \setminus E) \cup (P \cap E)$$

would be non-piecewise-syndetic. Their union would have the same property, contradicting the assumption on P.

For the final assertion, positive integer dilations preserve piecewise syndeticity. To see this directly, suppose $F + A$ is thick for a finite $F \subseteq \mathbb{Z}$. If it contains $\{b, \ldots, b+N-1\}$, then

$$\lambda A + \lambda F + \{0, 1, \ldots, \lambda - 1\}$$

contains the interval $\{\lambda b, \dots, \lambda(b+N)-1\}$. Thus λA is piecewise syndetic. If $E^{(\lambda)}$ were piecewise syndetic, so would be $\lambda E^{(\lambda)} \subseteq E$, which is impossible. □

In particular, if $\mathcal{H}$ is an allowed host, then so is every rescaled host

$$\mathcal{H}^{(\lambda)} = \{d \in \mathbb{N} : \lambda d \in \mathcal{H}\}. \tag{2.2}$$

Its complement is $(\mathbb{N} \setminus \mathcal{H})^{(\lambda)}$. This is the elementary fact that makes localization compatible with the integer scales used later.

2.3 Basic nilsystem facts

We use the following standard facts about niltranslations; see [5].

1. The orbit closure of a point under a niltranslation is a minimal nilsystem. Its forward orbit is dense in that orbit closure.
2. Such an orbit closure is a finite union of connected closed subnilmanifolds. For a translation, its connected components are cyclically permuted.
3. Products and powers of niltranslations are again niltranslations.

All topological systems below are compact metric systems whose transformations are homeomorphisms. For a minimal system (X, T), write

$$\pi : X \longrightarrow Z$$

for its maximal infinite-step pronilfactor. The space Z is an inverse limit of finite-dimensional minimal nilsystems, and every nilsystem factor of X factors through Z. Every nonempty open subset of Z contains the full inverse image of a nonempty open set in a finite nilsystem factor. For clarity, an inverse-limit cylinder may initially involve finitely many coordinates; passing to a later common coordinate combines those conditions into one open set. Surjectivity of the finite-coordinate projections guarantees that this open set can be chosen nonempty.

2.4 The return-time theorem and its exact use

For open sets $U_0, \dots, U_{h-1}$ in a system X, define

$$R_h^X(U_0, \dots, U_{h-1}) = \left\{ d \in \mathbb{Z} : \bigcap_{j=0}^{h-1} T^{-jd} U_j \neq \varnothing \right\}.$$

Write $R_h^X(U)$ when all the sets equal U. The integer d is a difference of orbit times, not a starting position.

Theorem 2.5 (Glasscock–Koutsogiannis–Le–Moreira–Richter–Robertson, linear form)**.** *Let (X, T) be minimal and invertible and let $\pi : X \to Z$ be its maximal infinite-step pronilfactor. For nonempty open $U_0, \dots, U_{h-1} \subseteq X$, the set*

$$R_h^Z\big((\pi U_0)^\circ, \dots, (\pi U_{h-1})^\circ\big) \setminus R_h^X(U_0, \dots, U_{h-1}) \tag{2.3}$$

is not piecewise syndetic in $\mathbb{Z}$.

This is Theorem A of [3], applied to the essentially distinct polynomials $0, n, \dots, (h-1)n$. Their pairwise differences are nonconstant, as required. In particular, that theorem allows the zero polynomial in this tuple. We retain the interiors in the statement; we will not assume that images of arbitrary open sets under π are open.

Lemma 2.6 (Lifting while excluding prescribed differences)**.** *Let $U \subseteq X$ be nonempty and open and suppose $\pi(U) = W$ is open in Z. Let $P \subseteq R_h^Z(W) \cap \mathbb{N}$ be piecewise syndetic and let $E \subseteq \mathbb{Z}$ be not piecewise syndetic. Then*

$$(P \setminus E) \cap R_h^X(U)$$

is piecewise syndetic, and in particular is nonempty.

Proof. Because $\pi(U) = W$ is open, Theorem 2.5 says that

$$F = R_h^Z(W) \setminus R_h^X(U)$$

is not piecewise syndetic. The set $E \cup F$ has the same property by Lemma 2.4. That lemma therefore makes $P \setminus (E \cup F)$ piecewise syndetic. Since $P \subseteq R_h^Z(W)$, this set equals the one in the assertion. □

The strength here is not merely the existence of a lifted difference. It is that an independently prescribed non-piecewise-syndetic excluded set can be removed as well. We do not replace "not piecewise syndetic" by "zero upper Banach density" for the exceptional set in Theorem 2.5; that stronger statement is not being assumed.

3 The finite-stage construction

Fix a host $\mathcal{H} \subseteq \mathbb{N}$ whose complement $E = \mathbb{N} \setminus \mathcal{H}$ is not piecewise syndetic. Fix $s \geq 1$ and an old set $B \subseteq \mathcal{H}$ with $\mathcal{P}_s$. Suppose also that

$$\chi : \mathbb{N} \to \{0, \dots, s\}$$

has no monochromatic B-progression of some length $H \geq 2$. The letter H without calligraphic type denotes this old forbidden length; $\mathcal{H}$ denotes the host. We want to add one color to the positive recurrence statement and one color to the obstructing palette. Put

$$u = s + 1, \qquad q = u + 1 = s + 2, \qquad Q = 100q, \qquad \delta = Q^{-1}, \qquad \eta = (10^6 Q^2)^{-1}. \tag{3.1}$$

The torus dimension M will be chosen later. Fix real numbers $\alpha_1, \dots, \alpha_M$ such that $1, \alpha_1, \dots, \alpha_M$ are linearly independent over $\mathbb{Q}$. For example, one can use the square roots of distinct primes. Write $\alpha = (\alpha_1, \dots, \alpha_M)$.

For $v \in \mathbb{T}^M$, use centered representatives $\theta_j \in [-1/2, 1/2)$ for its coordinates. Let O_M consist of those v for which some index i satisfies

$$|\theta_i + \delta| < \eta, \qquad |\theta_j| < \eta \quad (j \neq i), \qquad \sum_{j \neq i} |\theta_j| < 2\delta. \tag{3.2}$$

All these coordinates are away from the endpoints of the centered interval, so O_M is open in $\mathbb{T}^M$. Define

$$G_M = \{d \in \mathbb{N} : d\alpha \in O_M\}, \tag{3.3}$$

$$B_M = \left\{ d \in B : \sum_{j=1}^{M} \|d\alpha_j\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{4\pi q H} \right\}, \tag{3.4}$$

$$S_M = S_M(\mathcal{H}, B) = (G_M \cap \mathcal{H}) \cup B_M. \tag{3.5}$$

Because $B \subseteq \mathcal{H}$, the entire set S_M lies in $\mathcal{H}$. Only the new torus directions need to be explicitly intersected with the host.

A new direction has one coordinate of size approximately $-1/Q$, while its other coordinates are individually much smaller and have small total absolute size. An old direction in B_M moves the entire torus only slightly. The same coloring will handle these two types of motion.

Proposition 3.1 (Localized finite-stage step)**.** *Let $h \geq 2$ and let $C \geq 0$ bound the nilmanifold dimension. For all sufficiently large M, depending only on h, C, q, the following hold.*

1. *Every coloring with at most $s+1$ colors of a return-time set $N_T(y, V)$ in a nilsystem of dimension at most C contains a monochromatic h-term progression with difference in S_M.*
2. *There is a q-coloring κ_M of $\mathbb{N}$ with no monochromatic S_M-progression of length*

$$L = \max\{2Q, H\} = \max\{200q, H\}. \tag{3.6}$$

The bound on M is independent of the host and of B. The length L is independent of M, h, C and of the host.

For the initial step the same conclusions hold with $s = 0$, $B = \varnothing$, $H = 2$, $u = 1$, $q = 2$, and a constant input coloring χ. No assumption $\mathcal{P}_0$ is made.

Section 4 produces a syndetic supply of candidate torus directions compatible with the nilsystem window. Section 5 constructs the obstruction even for the larger set $G_M \cup B_M$. Section 6 uses Theorem 2.5 to retain a suitable difference in the host.

4 Why the torus has enough useful directions

4.1 A linear-algebra lemma

The following elementary fact explains why increasing M helps. A subspace of small codimension cannot exclude all vectors having one distinguished coordinate and many individually small correction coordinates.

Lemma 4.1. *Fix $b \geq 0$ and the positive numbers δ, η from (3.1). For all sufficiently large M, every real linear subspace $V \subseteq \mathbb{R}^M$ of codimension at most b contains a vector θ such that, for some i,*

$$\theta_i = -\delta, \qquad \max_{j \neq i} |\theta_j| < \eta, \qquad \sum_{j \neq i} |\theta_j| < 2\delta.$$

Consequently every connected subtorus of $\mathbb{T}^M$ of codimension at most b meets O_M.

Proof. Let $E = \mathbb{R}^M/V$ and let v_j be the image of the jth coordinate vector in E. Write $r = \dim E \leq b$. If $r = 0$, take $\theta = -\delta e_i$ and there is nothing more to prove. We may therefore assume $1 \leq r \leq b$.

Normalize the columns. Among all bases of E formed from r of the vectors v_j, choose one with maximal absolute determinant, computed in any fixed initial coordinates. Use these selected vectors as a new basis. Every v_j now has all coordinates in $[-1,1]$. Indeed, Cramer's rule expresses each coordinate as a determinant obtained by replacing one selected basis column, divided by the selected maximal determinant. Its absolute value cannot exceed one. This normalization is what makes the eventual bound independent of the particular subspace V.

Find many close columns. Put

$$\beta = \eta/\delta, \qquad R = \lceil 8/\beta \rceil, \qquad \tau_0 = \frac{\beta}{8b}, \qquad K = \left\lceil \frac{16b}{\beta} \right\rceil.$$

Here $0 < \beta < 1$. Partition $[-1,1]^r$ into at most $K^r \le K^b$ boxes of side length at most τ_0. If $M > RK^b$, one box contains at least $R+1$ indexed columns. Repeated vectors cause no problem; the indices, rather than the vector values, are what we count. Call one column v_i, let I be the indices of R other columns in that box, and put

$$\bar{v} = \frac{1}{R} \sum_{j \in I} v_j.$$

Each coordinate of $v_i - \bar{v}$ in our selected basis has absolute value at most τ_0. Consequently

$$v_i - \bar{v} = \sum_j c_j v_j, \qquad \sum_j |c_j| \le r\tau_0 \le \beta/8, \qquad \max_j |c_j| \le \tau_0,$$

where the c_j are zero outside the selected basis indices.

Correct the average exactly. Move $c_i v_i$ to the left and divide by $1 - c_i \ge 1 - \beta/8 > 0$. This gives

$$v_i = \sum_{j \ne i} t_j v_j, \qquad t_j = \frac{R^{-1} 1_I(j) + c_j}{1 - c_i}.$$

Using $R^{-1} \le \beta/8$ and $\tau_0 \le \beta/8$, we obtain

$$\sum_{j \ne i} |t_j| \le \frac{1 + \beta/8}{1 - \beta/8} \le \frac{9}{7} < 2, \qquad \max_{j \ne i} |t_j| \le \frac{\beta/4}{1 - \beta/8} \le \frac{2\beta}{7} < \beta.$$

Set $\theta_i = -\delta$ and $\theta_j = \delta t_j$ for $j \ne i$. Its image in E is zero, so $\theta \in V$, and the required bounds follow.

For the last assertion, apply the subspace statement to the Lie algebra of a connected subtorus and reduce θ modulo $\mathbb{Z}^M$. The coordinates just constructed satisfy (3.2). □

4.2 A syndetic supply of return steps with the desired torus motion

Lemma 4.2. *Fix $h \ge 2$ and $C \ge 0$. Choose M large enough for Lemma 4.1 with $b = hC$. For every nilsystem (Y,T) of dimension at most C, every $y \in Y$, and every open neighborhood V of y, the set*

$$P_h(y,V) = \{d \in \mathbb{N} : d\alpha \in O_M,\ T^{jd} y \in V \ (1 \le j \le h)\} \tag{4.1}$$

is syndetic in $\mathbb{N}$. In particular it is piecewise syndetic as a subset of $\mathbb{Z}$.

Proof. Consider the orbit of $(y, \dots, y, 0)$ in $Y^h \times \mathbb{T}^M$ under

$$(T, T^2, \dots, T^h, R_\alpha).$$

Let K be its orbit closure and let K_0 be the connected component containing the starting point. By the nilsystem facts above, K is minimal and is a finite union of connected closed subnilmanifolds.

The projection of K onto $\mathbb{T}^M$ is the whole torus, because $1, \alpha_1, \dots, \alpha_M$ are rationally independent. The projections of the finitely many components of K are translates of the same connected subtorus. If that subtorus were proper, finitely many of its cosets could not cover $\mathbb{T}^M$. For example, each such coset has Haar measure zero. Thus K_0 projects onto all of $\mathbb{T}^M$.

Present Y with a discrete lattice, so that the dimension of its ambient Lie group is $\dim Y \le C$. Write

$$K_0 = L \cdot (y, \dots, y, 0)$$

for a connected closed subgroup L of the product ambient group, and let $\mathfrak{l} = \mathrm{Lie}(L)$. Its projection onto the M torus directions is surjective. In particular $\dim \mathfrak{l} \ge M$. The first h factors together have Lie algebra dimension at most hC, so the kernel of their projection from $\mathfrak{l}$ has dimension at least $M - hC$. Equivalently, the pure-torus subspace

$$E_0 = \{v \in \mathbb{R}^M : (0, \dots, 0, v) \in \mathfrak{l}\}$$

has dimension at least $M - hC$.

Exponentiate E_0 in the torus and take its closure H_0. This is a connected subtorus of codimension at most hC. The corresponding group elements fix every Y-coordinate of the starting point. Since K_0 is closed, we obtain

$$\{(y, \dots, y)\} \times H_0 \subseteq K_0.$$

Lemma 4.1 gives $H_0 \cap O_M \neq \varnothing$. Consequently the relatively open set

$$K \cap (V^h \times O_M)$$

is nonempty. Let S denote the product transformation on K and let $\mathcal{O} = K \cap (V^h \times O_M)$. Minimality implies that every forward orbit enters the nonempty open set $\mathcal{O}$. Consequently the sets $S^{-t}\mathcal{O}$, $t \ge 0$, cover K. By compactness, finitely many of them cover K; choose R larger than all their indices. For every $n \ge 1$, the point $S^n(y, \dots, y, 0)$ therefore enters $\mathcal{O}$ after at most $R - 1$ further iterates. Thus every interval $\{n, \dots, n + R - 1\}$ meets $P_h(y, V)$, proving the syndeticity assertion. No uniform bound on R is claimed. □

The useful feature is the dependence of M only on h, C, q. It does not depend on the translation, the point, the size of the open window, or any coloring. The small window may make the required return time very large, but it does not require a larger torus dimension.

Remark 4.3 (Why the set of candidates, not just one candidate, matters). The geometric argument has not changed when we prescribe a host. It first supplies the same syndetic set $P_h(y, V)$ independently of the host and the coloring. Only afterward do we exclude the host's complement and the exceptional lifting set. Lemma 2.4 guarantees that a piecewise syndetic set of candidates remains. Selecting one candidate before the lifting argument would not provide this freedom.

5 The local coloring: one moving empty color

We now prove the obstructing-coloring part of Proposition 3.1. This part works for every M and is entirely elementary. In fact the coloring blocks $G_M \cup B_M$, so it also blocks its subset $S_M = (G_M \cap \mathcal{H}) \cup B_M$.

5.1 The palette before the formula

There are u old input colors and $q = u + 1$ new output colors. At each integer position we will choose an injection of the old palette into the new palette. Thus exactly one new color is unused at that position. The unused color changes as the position changes.

We need two features. On a G_M-progression, the unused color must eventually run through all q possibilities, which prevents any fixed output color from appearing at every site. On a B_M-progression, the injection must change so little that a constant output still forces a constant input.

Here is the general family of injections. For $J \in \mathbb{Z}$ and $i \in \{0, \dots, u-1\}$, choose the unique integer

$$t \in \{J-u, J-u+1, \dots, J-1\} \quad \text{with} \quad t \equiv i \pmod{u},$$

and put

$$\iota_J(i) = t - 1 \pmod{q}. \tag{5.1}$$

Example 5.1 (Two input colors and three output colors)**.** For $u = 2$ and $q = 3$, the injections repeat after six steps.

$J \pmod 6$	Output of input 0	Output of input 1	Missing output
0	0	1	2
1	2	1	0
2	2	0	1
3	1	0	2
4	1	2	0
5	0	2	1

In moving down one row, one input moves into the previously empty output color. An output that occurs in both rows retains the same input label.

Lemma 5.2. *The map ι_J is injective and omits exactly the color $J-1 \pmod q$. Moreover,*

$$|J - J'| \le 1, \qquad \iota_J(i) = \iota_{J'}(i') \quad \Longrightarrow \quad i = i'. \tag{5.2}$$

Proof. The integer outputs before reduction modulo q are

$$J-u-1, J-u, \dots, J-2 = J-q, J-q+1, \dots, J-2.$$

They are $q-1$ consecutive integers, so they are distinct modulo q and omit $J-1$. When J is increased to $J+1$, the representative $J-u$ leaves the input interval and J enters it. These two representatives correspond to the same input because their difference is u. Every other input keeps its representative and therefore its output. The changing input moves from output $J \pmod q$ to output $J-1 \pmod q$, the previously missing color. Its old output is missing in the new chart. Thus every output present in both consecutive charts belongs to the same input in both. Together with injectivity within a chart, this proves (5.2). □

5.2 Choosing the chart at an integer

Let $\mathrm{e}(t) = \exp(2\pi i t)$ and define

$$\Phi(n) = \sum_{j=1}^{M} \mathrm{Re}(\mathrm{e}(n\alpha_j) - 1), \qquad J(n) = \lfloor q\Phi(n) \rfloor. \tag{5.3}$$

The new coloring is

$$\kappa_M(n) = \iota_{J(n)}(\chi(n)) \in \mathbb{Z}/q\mathbb{Z}. \tag{5.4}$$

We use the elementary inequality

$$|\mathrm{e}(t) - 1| \le 2\pi \|t\|_{\mathbb{R}/\mathbb{Z}}. \tag{5.5}$$

5.3 Old differences: the old obstruction is preserved

Let $d \in B_M$ and consider the first H terms of a progression with starting point b. For any $0 \le r, t < H$, the triangle inequality and (5.5) give

$$|q\Phi(b + td) - q\Phi(b + rd)| \le 2\pi q|t - r| \sum_{j=1}^{M} \|d\alpha_j\|_{\mathbb{R}/\mathbb{Z}} \tag{5.6}$$

$$< \frac{H - 1}{2H} < \frac{1}{2}. \tag{5.7}$$

This is a pairwise estimate, so all the real numbers $q\Phi(b + td)$ lie in an interval of length less than $1/2$. Their floors therefore lie in at most two consecutive charts.

If the output colors $\kappa_M(b + td)$ were all equal, Lemma 5.2 would force all input colors $\chi(b + td)$ to be equal. Because $d \in B$, this contradicts the old length-H obstruction. Thus no B_M-progression of length H is monochromatic for κ_M.

5.4 New differences: every output color becomes unavailable

Let $d \in G_M$ and let θ_j be the centered coordinates of $d\alpha$. Choose the distinguished coordinate i from (3.2). We will show that on every progression of length $2Q$, the integers $J(n)$ visit every residue class modulo q. Since chart J omits color $J - 1$, this will rule out monochromaticity independently of the input coloring χ.

First estimate: the chart moves at most one step at a time. For consecutive sites,

$$|q\Phi(b + (r + 1)d) - q\Phi(b + rd)| \le 2\pi q \sum_{j} |\theta_j| < 2\pi q(3/Q + \eta) < 1. \tag{5.8}$$

Consequently $|J(b + (r + 1)d) - J(b + rd)| \le 1$.

Second estimate: the real values nevertheless have a large range. Choose an integer $0 \le t < Q$ such that

$$\|b\alpha_i - t/Q\|_{\mathbb{R}/\mathbb{Z}} \le \frac{1}{2Q}.$$

Such a t exists because the Q points $0, 1/Q, \dots, (Q-1)/Q$ are evenly spaced around the circle. Write $\theta_i = -1/Q + \epsilon$, where $|\epsilon| < \eta$. Then

$$\|b\alpha_i + t\theta_i\|_{\mathbb{R}/\mathbb{Z}} \le \frac{1}{2Q} + Q\eta < \frac{1}{Q}.$$

Since $Q \ge 200$, the phase

$$z = \mathrm{e}(b\alpha_i + t\theta_i)$$

has real part greater than 0.99.

Set $m = Q/2$, which is an integer. The three indices $t, t+m, t+2m$ all lie in $\{0, \dots, 2Q-1\}$. Consider the second difference

$$D = \Phi(b + td) - 2\Phi(b + (t+m)d) + \Phi(b + (t+2m)d).$$

Its distinguished-coordinate contribution is

$$\mathrm{Re}(z(\mathrm{e}(m\theta_i) - 1)^2).$$

Since $m\theta_i = -1/2 + m\epsilon$, we have

$$|\mathrm{e}(m\theta_i) + 1| \le \pi Q\eta.$$

Write $\mathrm{e}(m\theta_i) = -1 + w$. Expanding $(-2+w)^2$ shows that the distinguished contribution exceeds

$$3.96 - 4\pi Q\eta - \pi^2 Q^2 \eta^2.$$

The remaining coordinates have total contribution of absolute value at most

$$\sum_{j \ne i} |\mathrm{e}(m\theta_j) - 1|^2 \le \pi^2 Q^2 \sum_{j \ne i} \theta_j^2 \tag{5.9}$$

$$\le \pi^2 Q^2 \left(\max_{j \ne i} |\theta_j| \right) \sum_{j \ne i} |\theta_j| \tag{5.10}$$

$$< 2\pi^2 Q\eta. \tag{5.11}$$

Our choice $\eta = (10^6 Q^2)^{-1}$ therefore gives $D > 3.9$. If R is the range of Φ on the $2Q$ sites, then $D \le 2R$. Hence

$$R > 1.95.$$

The range of $q\Phi$ exceeds $1.95q > q$. Together with (5.8), this forces the integer sequence J to visit at least q consecutive integers. Indeed, a sequence of integers whose adjacent changes have magnitude at most one must pass through every integer between its minimum and maximum. Thus every residue modulo q is visited, and every output color is absent at some site. No G_M-progression of length $2Q$ is monochromatic.

Combining the old and new differences proves the second assertion of Proposition 3.1, with $L = \max\{2Q, H\}$.

Remark 5.3 (Why the dimension disappears). The first-difference bound uses the total sum $\sum_j |\theta_j|$. The second-difference error uses the maximum small coordinate times the sum of the small coordinates. Neither bound contains a factor M. This is why the same forbidden length works even as the auxiliary torus dimension increases.

6 Forcing one additional color inside the host

We prove the positive assertion of Proposition 3.1. Fix h, C, choose M by Lemma 4.2, and let c be an $(s+1)$-coloring of $N_T(y, V)$ in a nilsystem of dimension at most C. Extend c to a two-sided sequence $\omega \in \{1, \dots, s+1\}^{\mathbb{Z}}$ in any way. The colors outside the prescribed positive return-time set will not enter the final progression.

6.1 The joined system and its factors

Let Y_0 be the orbit closure of y and put $V_0 = V \cap Y_0$. Let σ be the left shift, so $(\sigma\omega)(n) = \omega(n+1)$. Consider the point

$$(\omega, y) \in \{1, \dots, s+1\}^{\mathbb{Z}} \times Y_0$$

under the product transformation (σ, T). Its forward limit set is the nonempty compact invariant set

$$\bigcap_{N \geq 1} \overline{\{(\sigma^n \omega, T^n y) : n \geq N\}}.$$

Choose a nonempty minimal subsystem X of that limit set. Its projection onto Y_0 is a nonempty compact invariant subset of the minimal nilsystem Y_0, hence is all of Y_0. Writing T also for the transformation on X, factor that projection as

$$X \xrightarrow{\pi} Z \xrightarrow{\rho} Y_0,$$

where Z is the maximal infinite-step pronilfactor of X. This factorization is available because Y_0 is a nilsystem factor.

Let $C_1 \subseteq X$ be the clopen cylinder where the symbol at coordinate zero is the first color. Put

$$W = \rho^{-1}(V_0).$$

This is a nonempty open set in Z. We do not join the auxiliary torus to X at this point. The return-time theorem will control the difference directly, so there is no need to recover its torus displacement from two orbit coordinates.

There are two cases: either the first color projects over the entire factor window W, or a nonempty open part of that window has no points of the first color above it.

6.2 First case: the first color covers the factor window

Assume $W \subseteq \pi(C_1)$ and set

$$U = C_1 \cap \pi^{-1}(W).$$

Then U is nonempty and open, and

$$\pi(U) = W. \tag{6.1}$$

To check the reverse inclusion in this equality, take $z \in W$. The covering assumption supplies a point of C_1 above z, and that point automatically belongs to $\pi^{-1}(W)$. Thus the interiors in Theorem 2.5 cause no loss here.

Surjectivity of ρ gives

$$R_h^Z(W) = R_h^{Y_0}(V_0). \tag{6.2}$$

One inclusion follows by projecting a configuration. For the other, lift its starting point arbitrarily through ρ; equivariance puts every lifted iterate in the full inverse image W.

Apply Lemma 4.2 to the original pointed nilsystem (Y, T, y) and its window V. Its syndetic set $P = P_h(y, V)$ consists of positive differences in G_M. Since $y \in V$ and all its relevant iterates remain in Y_0, it also satisfies

$$P \subseteq R_h^{Y_0}(V_0) = R_h^Z(W).$$

There is no need to apply the geometric lemma to a larger factor or to a nilsystem presentation of Y_0.

Let $E = \mathbb{N} \setminus \mathcal{H}$. Lemma 2.6, with this excluded set, shows that

$$P \cap \mathcal{H} \cap R_h^X(U) \tag{6.3}$$

is piecewise syndetic. Choose d in that set. Then $d \in G_M \cap \mathcal{H} \subseteq S_M$, and there is $x \in X$ such that

$$x, T^d x, \dots, T^{(h-1)d} x \in U.$$

These points all have the first color and all project to V_0. Notice that we have obtained a piecewise syndetic set of usable differences in this case, not only one. There is also a stronger relative statement:

$$P \setminus (\mathcal{H} \cap R_h^X(U)) \quad \text{is not piecewise syndetic.} \tag{6.4}$$

Indeed this loss is contained in the union of E and the exceptional set in Theorem 2.5. Consequently every piecewise syndetic subset of P still has a piecewise syndetic intersection with the usable differences. A conclusion giving merely some syndetic set of lifted differences does not give this control relative to every such subset of P.

6.3 Second case: remove the first color

Assume $W \not\subseteq \pi(C_1)$. The set $\pi(C_1)$ is compact, hence closed, because C_1 is compact. Therefore $W \setminus \pi(C_1)$ is nonempty and open. Choose a finite nilsystem factor $\zeta : Z \to Z'$ and a nonempty open set $V' \subseteq Z'$ such that

$$\zeta^{-1}(V') \subseteq W \setminus \pi(C_1). \tag{6.5}$$

Every point of X above this cylinder has one of the remaining s colors, and its Y_0-coordinate belongs to V_0.

Choose $x_0 \in X$ with $z' = \zeta\pi(x_0) \in V'$. Choose an open neighborhood O of 0 in $\mathbb{T}^M$ so small that

$$\sum_{j=1}^{M} \|v_j\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{4\pi q H} \quad \text{whenever } v \in O - O. \tag{6.6}$$

For example, a sufficiently small product of centered intervals has this property, by the triangle inequality. Consider visits of $(z', 0)$ to $V' \times O$ under the product niltranslation (T, R_α). Color a visiting time n by the symbol at coordinate zero of $T^n x_0$. By (6.5), this uses at most s colors.

Property $\mathcal{P}_s$ of B supplies positive times

$$b, b + d, \dots, b + (h-1)d$$

which are such visits, all have the same color, and have $d \in B$. The first two torus visits show that $d\alpha \in O - O$. Consequently (6.6) puts d in B_M. Since $B \subseteq \mathcal{H}$ was chosen at the start of this finite stage, the difference also lies in the host. The corresponding configuration in X has one color and projects into V_0.

6.4 Returning to the original coloring

In either case there is a point $x \in X$ and a fixed positive difference $d \in S_M$ such that

$$x, T^d x, \dots, T^{(h-1)d} x$$

have one color and project into V_0. Specifying that color at the finitely many coordinates $0, d, \dots, (h-1)d$ is a clopen symbolic condition. Requiring the corresponding Y_0-coordinates to lie in V_0 is an open condition. Thus the configuration is witnessed by an open neighborhood of x in the ambient joined system.

Because X lies in the forward limit set of (ω, y), this neighborhood is visited at arbitrarily late positive times a along the original joined orbit. At each such time,

$$a, a+d, \dots, a+(h-1)d$$

are positive visits of y to V and have equal original colors. All their colors were part of the original coloring of $N_T(y, V)$, not of its arbitrary extension. This proves the positive assertion of Proposition 3.1.

For $s = 0$, the symbolic sequence has only one color, so $C_1 = X$ and the second case cannot occur. The first case proves the initial assertion without any hypothesis $\mathcal{P}_0$.

Remark 6.1 (Uniformity and its limits). The geometric dimension bound is applied only to the original nilmanifold of dimension at most C. The finite nilfactor Z' in the second case can have larger dimension; this is harmless because $\mathcal{P}_s$ already concerns every finite-dimensional nilsystem. Thus M depends only on h, C, q, not on the host, the coloring, or the symbolic extension. The piecewise-syndetic conclusion (6.3) is specific to the first case. We do not claim that the full set of successful differences is piecewise syndetic in the second case, or after the final packet assembly.

Remark 6.2 (A simplification provided by the stronger theorem). Compared with the Qiu-only argument, the main proof no longer needs an initial joining with $\mathbb{T}^M$, open phase targets P_0, P_1, or a separate device for obtaining a positive lifted difference. The candidate set P already consists of positive integers with the required torus motion, and Theorem 2.5 allows us to choose the lifted difference directly from it. The auxiliary torus is still used in the geometry, the coloring, and the second-case restriction to B_M.

7 Finite packets and the passage to one set of integers

Proposition 3.1 provides a useful set $S_M(\mathcal{H}, B)$ for each fixed target length, dimension bound, host, and old set. Taking a union of these sets would not preserve the obstruction: the colorings κ_M change with M. Instead, we extract a finite packet from each stage and place those packets at carefully chosen integer scales.

There are two separate tasks. First, a finite packet must handle an entire compact family of recurrence tests. Second, the local obstructing colorings must be assembled without increasing the number of colors. We treat these tasks in that order.

7.1 A countable list of compact tests

The following parametrization lemma is proved in Appendix A. Its useful simplification is that every test starts at an identity coset. There is no variable starting-point parameter.

Lemma 7.1 (Countable tests)**.** *There is a countable list of data*

$$\mathcal{T} = (N, \Lambda, A, m, K, V) \tag{7.1}$$

with the following properties. The group N is connected, simply connected and nilpotent, Λ is a lattice, and A is a lattice-preserving unipotent automorphism. The integer m is positive, $K \subseteq N$ is compact, and V is an open neighborhood of $o = e\Lambda$ in N/Λ. For $b \in K$, let

$$R_b(z\Lambda) = bA(z)\Lambda.$$

This is a niltranslation in a possibly disconnected presentation, and every power R_b^e is a niltranslation on a nilmanifold of the same dimension $\dim N$*.*

Every pointed nilsystem (Y, T, y) with an open neighborhood U of y is represented by one of these tests after restricting to multiples of its integer m: there are $b \in K$ and an identification $\phi : N/\Lambda \to Y_y$, where Y_y is the component of y, such that

$$\phi(o) = y, \qquad \phi \circ R_b = T^m \circ \phi, \qquad \phi(V) \subseteq U.$$

Fix an enumeration $\mathcal{T}_1, \mathcal{T}_2, \dots$ of these tests. At every stage we also include the one-point nilsystem with its whole space as window and $m = 1$. This ensures that the finite packets extracted below are nonempty.

7.2 What is fixed before a finite packet is chosen

Fix the desired global host $\mathcal{H}$ and one induction level s. For $s \geq 1$, assume the full statement (I_s) in (2.1), including its uniform forbidden length H_s. Set

$$H = H_s, \qquad q = s + 2, \qquad L = \max\{200q, H\}.$$

For $s = 0$, use $H = 2$, $q = 2$, and empty old sets at every stage. The numbers s, q, H, L are fixed throughout this assembly. The old recurrence sets need not be fixed.

Start with $\lambda_1 = 1$. At stage j, the scale λ_j has already been chosen. Define the local host

$$\mathcal{H}_j = \{d \in \mathbb{N} : \lambda_j d \in \mathcal{H}\}. \tag{7.2}$$

Its complement is not piecewise syndetic by Lemma 2.4. Apply (I_s) to this host to choose

$$B_j \subseteq \mathcal{H}_j, \qquad B_j \text{ has } \mathcal{P}_s, \tag{7.3}$$

and a coloring $\chi_j : \mathbb{N} \to \mathbb{Z}/(s+1)\mathbb{Z}$ blocking all B_j-progressions of length H_s. Identify its colors with $\{0, \dots, s\}$ when applying the moving-palette formula. For $s = 0$, instead take $B_j = \varnothing$ and χ_j constant.

Set $h_j = j + 2$. From the first j compact tests, retain those whose integer m divides λ_j, and include the one-point test. Let C_j be the largest dimension of those tests. Choose M_j sufficiently large for Proposition 3.1, and apply it with host $\mathcal{H}_j$, old set B_j, and input coloring χ_j. Write

$$S_j = S_{M_j}(\mathcal{H}_j, B_j) \subseteq \mathcal{H}_j \tag{7.4}$$

and let κ_j be the resulting local q-coloring. Every κ_j blocks its set S_j at the same length L.

For a retained test, use the transformation

$$R_b^{e_j}, \qquad e_j = \lambda_j/m.$$

Its exponent is fixed before compactness is applied. Although the exponent may be large, the nilmanifold dimension is unchanged.

Remark 7.2 (Why the old set must be allowed to vary). Reusing a single old set B would require it to supply differences avoiding every excluded pullback $(\mathbb{N} \setminus \mathcal{H})^{(\lambda_j)}$. The non-piecewise-syndeticity of each individual pullback does not imply that their countable union is non-piecewise-syndetic. Nor does $\mathcal{P}_s$ justify deleting such sets from a fixed B. The correct induction is therefore uniform over hosts: choose B_j afresh inside $\mathcal{H}_j$. There is no circularity, because each such choice uses only the already established color level s, not the level $s+1$ under construction.

7.3 Why a finite packet works uniformly

Fix one retained test (N, Λ, A, m, K, V). Consider the compact parameter space

$$K \times \{1, \dots, s+1\}^{\mathbb{Z}}. \tag{7.5}$$

The first coordinate chooses the translation parameter b; the second chooses an arbitrary coloring ω of the integer times. For fixed positive integers a, d, with $d \in S_{M_j}$, impose the conditions

$$\omega(a) = \omega(a+d) = \cdots = \omega(a+(h_j-1)d), \qquad (R_b^{e_j})^{a+td} o \in V \quad (0 \le t < h_j). \tag{7.6}$$

They define an open subset of (7.5). The color equalities inspect only finitely many coordinates, so they are clopen conditions. Each fixed iterate depends continuously on b, so the window conditions are open.

Proposition 3.1 says that every parameter in (7.5) satisfies (7.6) for some positive a and some $d \in S_{M_j}$. Thus these open sets cover the compact parameter space. Choose a finite subcover and retain the finitely many differences occurring in it. Repeat for the finitely many tests at this stage, and let F_j be the union of the retained differences. Then

$$\varnothing \neq F_j \subseteq S_j \subseteq \mathcal{H}_j, \qquad |F_j| < \infty. \tag{7.7}$$

The packet F_j works for every translation parameter and every coloring in every retained test. Because $F_j \subseteq S_j$, the same local coloring κ_j blocks F_j-progressions of length L.

The minimal symbolic extension and the finite nilfactor appearing in the proof of Proposition 3.1 can depend on the coloring. They are not part of the parameter space (7.5). Compactness is applied only after a finite configuration has been obtained in the original nilsystem and coloring.

7.4 Choosing the next scale

After choosing F_j, choose an integer radix r_j such that

$$r_j > 2L \max F_j, \qquad (j+1)! \mid r_j, \tag{7.8}$$

and put

$$\lambda_{j+1} = r_j \lambda_j. \tag{7.9}$$

For example, $r_j = (j+1)!(2L \max F_j + 1)$ is an admissible choice. By induction,

$$j! \mid \lambda_j \qquad (j \ge 1). \tag{7.10}$$

Define the new set by

$$D = \bigcup_{j \ge 1} \lambda_j F_j. \tag{7.11}$$

The order of choices is important:

$$\lambda_j \longrightarrow \mathcal{H}_j \longrightarrow (B_j, \chi_j) \longrightarrow S_j \longrightarrow F_j \longrightarrow (r_j, \lambda_{j+1}).$$

The retained tests and the choice of M_j occur before S_j is formed. There is no need to choose a radix before knowing how large the finite packet is. The finite packet has already been selected inside $\mathcal{H}_j$, so choosing a larger later radix cannot spoil membership of $\lambda_j F_j$ in the global host.

7.5 The set D has the stronger recurrence property

Proposition 7.3. *The set D in (7.11) is contained in $\mathcal{H}$ and has property $\mathcal{P}_{s+1}$. For the initial step $s = 0$, it has $\mathcal{P}_1$.*

Proof. For each j, the inclusions $F_j \subseteq \mathcal{H}_j$ and the definition (7.2) imply $\lambda_j F_j \subseteq \mathcal{H}$. Taking the union shows $D \subseteq \mathcal{H}$.

Fix a pointed nilsystem (Y, T, y), an open neighborhood U of y, a coloring of $N_T(y, U)$ with $s+1$ colors, and a desired length h. Use Lemma 7.1 to select a test (N, Λ, A, m, K, V) and a parameter $b \in K$ representing the component return map and a window inside U. Choose j large enough that this test is among the first j tests, $j \geq m$, and $h_j \geq h$. By (7.10), $m \mid \lambda_j$, so the test is retained.

For $n \geq 1$, read the original coloring at time $\lambda_j n$. Whenever

$$(R_b^{\lambda_j/m})^n o \in V,$$

the original time $\lambda_j n$ is a visit to U, so its color is defined. Extend this coloring arbitrarily to all integer indices n. The defining property of F_j gives a monochromatic progression

$$a, a+d, \ldots, a+(h_j-1)d, \qquad a > 0, \quad d \in F_j,$$

all of whose iterates under $R_b^{\lambda_j/m}$ visit V. In the original system these are the positive times

$$\lambda_j a, \lambda_j a + \lambda_j d, \ldots, \lambda_j a + (h_j - 1)\lambda_j d.$$

They visit U, have the same original color, and have common difference $\lambda_j d \in D$. Taking the first h terms proves the assertion. The proof is identical for the initial one-color stage. □

The proof uses only the recurrence guarantee of each packet, not any compatibility between the old sets B_j. The obstruction also needs no such compatibility: it will use only the common output group $\mathbb{Z}/q\mathbb{Z}$ and the common forbidden length L.

8 Combining the local colorings by digits

We now construct a single q-coloring that blocks the whole union (7.11). This is a general device: it uses only the finite packets, their local colorings, and the radix inequality in (7.8).

8.1 Mixed-radix notation

Every nonnegative integer has a unique expansion

$$n = \sum_{j \geq 1} z_j(n)\lambda_j, \qquad 0 \leq z_j(n) < r_j. \tag{8.1}$$

Only finitely many digits are nonzero. This is the usual positional expansion, except that the radix at place j is r_j rather than a fixed radix such as ten. Equivalently,

$$z_j(n) = \left\lfloor \frac{n}{\lambda_j} \right\rfloor \pmod{r_j}, \qquad 0 \leq z_j(n) < r_j.$$

Use the additive group $\mathbb{Z}/q\mathbb{Z}$ as the output palette and define

$$c(n) = \sum_{j \geq 1} \bigl(\kappa_j(1 + z_j(n)) - \kappa_j(1)\bigr) \pmod{q}. \tag{8.2}$$

The subtraction makes a zero digit contribute zero. Therefore the sum is finite for each n, and it uses exactly the same q output labels as the local colorings. The shift by one makes all arguments of κ_j positive.

8.2 Why carries do not destroy the obstruction

Proposition 8.1 (Digit assembly)**.** *The coloring* (8.2) *has no monochromatic D-progression of length $2L$.*

Proof. Suppose, to the contrary, that a progression of length $2L$ and common difference $d \in D$ is monochromatic. Choose j and $e \in F_j$ such that $d = \lambda_j e$. Write its sites as

$$n_t = n_0 + t\lambda_j e, \qquad 0 \leq t < 2L.$$

The lower digits are fixed. All n_t have the same residue modulo λ_j. Thus every digit in a place below j is constant on the progression.

The higher prefix changes at most once. The total span is

$$n_{2L-1} - n_0 = (2L-1)\lambda_j e < \lambda_j r_j = \lambda_{j+1}.$$

Consequently the integer

$$p_t = \left\lfloor \frac{n_t}{\lambda_{j+1}} \right\rfloor$$

has at most two values and changes at most once. The $2L$ sites split into at most two consecutive runs on which p_t is constant. One of those runs has at least L sites. On this run all digits above place j are constant. This argument includes a carry that propagates through several higher places, because p_t records all the higher digits at once.

The remaining digit sees the local obstruction. On the chosen run, both the lower residue and the higher prefix are fixed. Therefore the jth digit increases by exactly e at each step, with no wrap inside that run. All summands in (8.2) except the jth are constant. Monochromaticity of the global coloring forces

$$\kappa_j(1+z),\ \kappa_j(1+z+e),\ \ldots,\kappa_j(1+z+(L-1)e)$$

to be constant for some nonnegative digit z. This is a forbidden local progression of length L, since $e \in F_j \subseteq S_j$. □

Example 8.2 (One carry is harmless)**.** For an illustration, take $L = 4$, $e = 3$, and a radix $r_j = 32 > 2Le$. Eight successive jth digits might be

$$26,\ 29,\ 0,\ 3,\ 6,\ 9,\ 12,\ 15.$$

The higher prefix changes between the second and third sites and stays fixed afterward. The final six sites contain a four-term run on which the local digit is an ordinary arithmetic progression of difference 3. The proof uses exactly this observation for arbitrary radices and packet differences.

Remark 8.3 (Why finite extraction is essential)*.* The choice $r_j > 2L \max F_j$ requires a finite upper bound for the differences used at one digit place. For an entire infinite set S_j, no finite radix would provide this control. Finite extraction is therefore part of the mechanism that preserves the obstructing palette.

9 Completing the localized induction

We now collect the construction and verify the numerical bound. At each color level the countable test list is fixed as in Lemma 7.1, and the packet procedure is run for the prescribed host.

9.1 The base case for every host

Let $\mathcal{H} \subseteq \mathbb{N}$ have non-piecewise-syndetic complement. Use

$$s = 0, \qquad H_0 = 2, \qquad q = 2, \qquad B_j = \varnothing, \quad \chi_j(n) = 0$$

at every packet stage. The local hosts are still $\mathcal{H}_j$ from (7.2). The initial case of Proposition 3.1 forces one-color recurrence inside each bounded-dimensional compact test family with differences in $S_j \subseteq \mathcal{H}_j$. The same local sets have two-color obstructions of length

$$L_1 = \max\{200 \cdot 2, 2\} = 400.$$

Finite extraction and Proposition 7.3 give $D_{1,\mathcal{H}} \subseteq \mathcal{H}$ with $\mathcal{P}_1$. Proposition 8.1 gives a two-color obstruction of length

$$H_1 = 2L_1 = 800.$$

The argument works for every permitted host, with the same value H_1. This establishes (I_1), not just the case of the host $\mathbb{N}$.

9.2 The induction step and its uniformity

Assume (I_s) for some $s \geq 1$. Fix an arbitrary permitted global host $\mathcal{H}$. At stage j, Lemma 2.4 makes $\mathcal{H}_j$ a permitted host, and (I_s) supplies $B_j \subseteq \mathcal{H}_j$ with $\mathcal{P}_s$ and an $(s+1)$-color obstruction of length H_s. Apply the finite-stage construction with $q = s+2$. All stages have the common local forbidden length

$$L = \max\{200(s+2), H_s\}.$$

The finite-packet argument gives a set $D_{s+1,\mathcal{H}} \subseteq \mathcal{H}$ with $\mathcal{P}_{s+1}$. The digit coloring uses $s+2$ colors and blocks length

$$H_{s+1} = 2\max\{200(s+2), H_s\}. \tag{9.1}$$

This bound is independent of the global host, all local hosts, and all choices of old sets. Hence (I_{s+1}) holds for every permitted host, completing the induction.

9.3 The advertised numerical bound

We have $H_1 = 800 = 400 \cdot 2$. For $s \geq 1$,

$$200(s+2) \leq 400 \cdot 2^s.$$

Thus if $H_s = 400 \cdot 2^s$, equation (9.1) gives

$$H_{s+1} = 2H_s = 400 \cdot 2^{s+1}.$$

The formula (1.1) follows for all $k \geq 1$. The auxiliary value $H_0 = 2$ is separate and is not obtained by substituting $k = 0$ into that formula.

9.4 Eventual divisibility and zero upper Banach density

For a set $A \subseteq \mathbb{N}$, define

$$d^*(A) = \limsup_{N\to\infty} \sup_{a\geq 0} \frac{|A \cap \{a+1, \ldots, a+N\}|}{N}.$$

We verify the additional sparsity conclusions directly from the assembled form $D = \bigcup_j \lambda_j F_j$.

Proposition 9.1. *For every $m \geq 1$, the set $D \setminus m\mathbb{N}$ is finite. Moreover $d^*(D) = 0$.*

Proof. Choose $J \geq m$. For every $j \geq J$, we have $m \mid j! \mid \lambda_j$. Thus every packet $\lambda_j F_j$ with $j \geq J$ lies in $m\mathbb{N}$. The earlier packets have finite union, proving the first assertion.

For a uniform density estimate, fix J and put

$$E_J = \bigcup_{j<J} \lambda_j F_j.$$

Since each later scale is a multiple of λ_J, we have

$$D \subseteq E_J \cup \lambda_J\mathbb{N}.$$

Every interval of length N therefore contains at most

$$|E_J| + \frac{N}{\lambda_J} + 1$$

points of D. Taking the supremum over interval locations and then the limit superior as $N \to \infty$ gives $d^*(D) \leq 1/\lambda_J$. Since $\lambda_J \geq J!$ tends to infinity, $d^*(D) = 0$. □

For reference, every piecewise syndetic subset of $\mathbb{N}$ has positive upper Banach density. If $F + A$ contains arbitrarily long intervals and F is finite, then on each such interval the sum of the counts in its finitely many translated copies of A is at least the interval length. One translate therefore has density at least $1/|F|$ on that interval. The associated intervals for A have the same length; bounded shifts at the origin do not affect the limiting bound. Consequently $d^*(A) \geq 1/|F| > 0$. This justifies the examples in Section 1 and shows that our final sets are not piecewise syndetic.

Property $\mathcal{P}_k$, the obstruction, and Proposition 9.1 prove all parts of Theorem 1.2.

9.5 The construction as a recipe

Here are the choices at one color level, with the two indices kept separate.

1. Fix the desired host $\mathcal{H}$ and the old color level s. Assume (I_s) when $s \geq 1$, or use the empty initial data when $s = 0$.
2. Fix $q = s + 2$, the old forbidden length H, and $L = \max\{200q, H\}$. Set $\lambda_1 = 1$.
3. At packet stage j, form $\mathcal{H}_j = \{d : \lambda_j d \in \mathcal{H}\}$. Choose the old set B_j and its input coloring inside this local host by (I_s).
4. Retain the compatible compact tests, with $h_j = j+2$ and maximum dimension C_j. Choose M_j and form $S_j = (G_{M_j} \cap \mathcal{H}_j) \cup (B_j)_{M_j}$. Here $(B_j)_{M_j}$ denotes the small-torus-motion restriction in (3.4).
5. Extract a nonempty finite packet $F_j \subseteq S_j$ by a finite subcover. The moving-palette formula supplies its local obstructing coloring κ_j.
6. Choose $r_j > 2L \max F_j$ divisible by $(j+1)!$, and put $\lambda_{j+1} = r_j \lambda_j$.
7. Set $D_{s+1,\mathcal{H}} = \bigcup_j \lambda_j F_j$ and use the digit coloring (8.2).

The host is enforced before each packet is chosen. Positive recurrence is carried by the packet tests. The obstruction is carried by the digit formula. Uniformity of the old forbidden length over all hosts is what lets these three requirements coexist.

10 A piecewise-syndetic criterion and the first separation

The return-time theorem also gives a useful general observation about the stronger induction property. It explains both the short first separation and why the full localized construction naturally produces non-piecewise-syndetic sets.

Proposition 10.1 (One additional color from piecewise syndeticity)**.** *Let $s \geq 1$. If $D \subseteq \mathbb{N}$ is piecewise syndetic and has $\mathcal{P}_s$, then D is $(s+1)$-large.*

Proof. Fix a coloring of $\mathbb{N}$ with $s+1$ colors and a desired length h. Extend it to a two-sided sequence and choose a minimal symbolic subsystem in its forward limit set, as in Section 6. Let C_1 be the first color cylinder and let $\pi : X \to Z$ be the maximal infinite-step pronilfactor.

If $\pi(C_1) = Z$, Theorem 2.5 gives

$$\mathbb{Z} \setminus R_h^X(C_1) \quad \text{not piecewise syndetic.}$$

It cannot contain the piecewise syndetic set D. Choose $d \in D \cap R_h^X(C_1)$. This gives a monochromatic length-h configuration with a positive difference in D.

If $\pi(C_1) \neq Z$, the nonempty open set $Z \setminus \pi(C_1)$ contains a finite nilfactor cylinder $\zeta^{-1}(V')$. Choose $x_0 \in X$ above a point $z' \in V'$. The times at which $T^n z'$ visits V' carry at most s colors from the orbit of x_0. Apply $\mathcal{P}_s$ of D to this nilsystem window to obtain a monochromatic length-h progression with difference in D.

In either case the finite configuration is specified by open conditions in the symbolic system. Because the subsystem lies in the forward limit set of the original sequence, the same configuration appears at a positive starting position in the original coloring. □

Remark 10.2. A set with $\mathcal{P}_s$ and an $(s+1)$-color obstruction cannot be piecewise syndetic, by Proposition 10.1. Thus the sparsity of the constructed $D_{s,\mathcal{H}}$ is compatible with a genuine structural restriction, not just an artifact of the scales. This does not prevent a different, merely s-large set from being syndetic.

Once a set E with $\mathcal{P}_1$ and a binary obstruction is available, the first strict separation has a short modular proof. One may take $E = D_{1,\mathbb{N}}$, without importing a separate construction.

Proposition 10.3. *Suppose E has $\mathcal{P}_1$ and a coloring $f : \mathbb{N} \to \{0,1\}$ has no monochromatic E-progression of length H. Then*

$$S = (\mathbb{N} \setminus 3\mathbb{N}) \cup 3E$$

is syndetic and 2-large, and a three-coloring blocks all its progressions of length $\max\{3, H\}$.

Proof. The set S is syndetic because it contains every positive integer not divisible by three. Lemma 2.2 shows that $3E$ has $\mathcal{P}_1$. Since the window property is upward closed, S also has $\mathcal{P}_1$. Proposition 10.1, with $s = 1$, proves that S is 2-large.

For the obstruction, write each positive integer uniquely as $3n + r$, with $r \in \{0,1,2\}$ and $n \geq 0$. On the three residue classes use, respectively, the ordered output palettes

$$(0,1), \qquad (1,2), \qquad (2,0),$$

choosing the first or second entry according as $f(n+1) = 0$ or 1. A difference not divisible by three visits all three residue classes in its first three terms. The sets of available colors $\{0,1\}, \{1,2\}, \{2,0\}$ have empty intersection, so such a progression cannot be monochromatic.

A difference in $S \cap 3\mathbb{N}$ has the form $3e$ with $e \in E$. It stays in one residue class, and dividing the changes in the indices by three gives a progression of quotient indices with difference e. Within a fixed residue class the palette map is injective, so constant output color would force constant input values of f. This is impossible on H terms. The two cases prove the stated obstruction length. □

The modular construction illustrates the missing-color idea without the dimension and compactness bookkeeping. It does not supply the stronger induction property $\mathcal{P}_2$ for S; in fact Proposition 10.1 and the three-color obstruction show that S cannot have $\mathcal{P}_2$. The full hierarchy therefore still needs the moving torus palette and the window-sensitive induction.

11 Comparison with Qiu's theorem

This section records the alternative dynamical input in detail. It is not needed for the proof of the localized theorem above. Its purpose is to explain why the almost one-to-one modification is

harmless, how Qiu gives the unrestricted hierarchy, and how an additional syndeticity fact recovers the localized hierarchy as well. This comparison distinguishes the amount of recurrence needed for the construction from the sharper relative exceptional-set information supplied by Theorem A.

11.1 The modification and the positive lifting lemma

Qiu's Theorem B [4] supplies minimal systems X^*, Z^* and factor maps

$$\tau : X^* \to X, \qquad \sigma : Z^* \to Z, \qquad \pi^* : X^* \to Z^*, \qquad \sigma\pi^* = \pi\tau, \tag{11.1}$$

where τ and σ are almost one-to-one. For open $V_0, \dots, V_{h-1} \subseteq X^*$ whose images under π^* have a common point, the theorem gives simultaneous visits along distinct nonconstant integer polynomials p_j, $1 \le j < h$, with $p_j(0) = 0$. The zeroth visit is to V_0 at time zero. This statement requires minimality, not total minimality.

An almost one-to-one map has singleton fibers over a dense set of base points. In the following proof, the singleton-fiber property is used for σ; only surjectivity and equivariance of τ are needed.

Lemma 11.1 (Positive recurrence lifting from Qiu)**.** *Let $U_0, \dots, U_{h-1}$ be open subsets of a minimal invertible system X. Suppose that for some $m \in \mathbb{N}$,*

$$I = \bigcap_{j=0}^{h-1} T^{-jm}(\pi(U_j))^\circ \neq \varnothing. \tag{11.2}$$

For each prescribed integer $a \ge 1$, there are $x \in X$ and $n \in \mathbb{Z}$ such that, with

$$d = m + an^2 > 0,$$

we have $T^{jd}x \in U_j$ for $0 \le j < h$.

Proof. The set I is nonempty and open. Choose $z \in I$ such that $\sigma^{-1}(z) = \{z^*\}$ is a singleton. This is possible because the singleton fibers occur over a dense subset of Z. For each j, set

$$V_j = \tau^{-1}(T^{-jm}U_j).$$

These are open subsets of X^*.

We claim that all their π^*-images contain z^*. Since $z \in T^{-jm}\pi(U_j)$, equivariance of π supplies $x_j \in T^{-jm}U_j$ with $\pi(x_j) = z$. Choose any lift $x_j^* \in \tau^{-1}(x_j)$. Then $x_j^* \in V_j$ and commutativity gives

$$\sigma(\pi^*(x_j^*)) = \pi(\tau(x_j^*)) = \pi(x_j) = z.$$

The singleton-fiber condition forces $\pi^*(x_j^*) = z^*$ for every j. Thus

$$z^* \in \bigcap_{j=0}^{h-1} \pi^*(V_j).$$

The lifts x_j^* need not coincide; the point is that their images coincide. This is exactly the intersection hypothesis in Qiu's theorem.

Apply that theorem with $p_j(n) = ajn^2$ for $1 \le j < h$. These polynomials are distinct, nonconstant, integer-valued, and vanish at zero. It gives $x^* \in X^*$ and $n \in \mathbb{Z}$ such that

$$x^* \in V_0, \qquad T^{ajn^2}x^* \in V_j \quad (1 \le j < h).$$

Let $x = \tau(x^*)$. The definition of V_j and equivariance of τ give

$$T^{j(m+an^2)}x \in U_j \qquad (0 \le j < h).$$

Since $m > 0$, the difference $m + an^2$ is positive even if Qiu's parameter n is zero or negative. It also satisfies $d \equiv m \pmod a$. □

The openness of I is the reason a singleton-fiber point can be chosen. An intersection consisting only of exceptional base points would not justify the same argument. We also make no claim that Z^* itself is a pronilsystem. All nilsystem geometry stays on Z and its finite factors; the modified spaces are used only to apply Qiu and project the resulting finite configuration back to X.

Remark 11.2 (An intrinsic formulation avoids the diagram). The six-author paper formulates a result equivalent to Qiu's Theorem B as its Theorem 1.1 and proves the equivalence in Theorem 3.9 [3]. That formulation says that nonemptiness of the factor return-time set for essentially distinct integer polynomials implies nonemptiness of the original return-time set. It is an existence statement, not Theorem A's exceptional-set statement.

From this formulation, Lemma 11.1 has a short proof. Use the polynomials

$$p_j(n) = j(m + an^2), \qquad 0 \le j < h.$$

Condition (11.2) witnesses a factor configuration at the parameter $n = 0$. The existence statement supplies an original-system configuration at some integer parameter, and its difference $m + an^2$ is positive. Thus a shorter exposition of the unrestricted proof can omit the modification diagram altogether by citing this equivalent formulation. The direct proof above is retained to make the role of almost one-to-one extensions explicit.

11.2 Recovering the unrestricted finite-stage argument

For this subsection only, take the host $\mathcal{H} = \mathbb{N}$. To replace Theorem 2.5 by Lemma 11.1 in the first case of Section 6, join the original colored sequence to the auxiliary torus as well as the nilsystem:

$$(\omega, y, 0) \in \{1, \dots, s+1\}^{\mathbb{Z}} \times Y_0 \times \mathbb{T}^M.$$

Choose a minimal subsystem of its forward limit set. The joint orbit closure of $(y, 0)$ under (T, R_α) is a minimal nilsystem, so the subsystem projects onto it and this projection factors through its maximal pronilfactor Z. There are therefore maps

$$\pi : X \to Z, \qquad \rho : Z \to Y_0, \qquad \gamma : Z \to \mathbb{T}^M, \qquad \gamma(Tz) = \gamma(z) + \alpha.$$

As before let $W = \rho^{-1}(V_0)$ and suppose $W \subseteq \pi(C_1)$.

Choose $z_0 \in Z$ with $\rho(z_0) = y$, and put $t_0 = \gamma(z_0)$. Lemma 4.2 supplies a positive $m \in G_M$ such that $T^{jm}y \in V_0$ for $0 \le j < h$. Only one such m is needed in this argument. Since $m\alpha \in O_M$ and subtraction on the torus is continuous, choose open neighborhoods

$$P_0 \ni t_0, \qquad P_1 \ni t_0 + m\alpha, \qquad P_1 - P_0 \subseteq O_M.$$

Put $P_j = \mathbb{T}^M$ for $j \ge 2$ and set

$$A_j = W \cap \gamma^{-1}(P_j), \qquad U_j = C_1 \cap \pi^{-1}(A_j).$$

The covering assumption gives $\pi(U_j) = A_j$, and these images are open. Moreover $T^{jm}z_0 \in A_j$ for every j, so (11.2) holds.

Apply Lemma 11.1 with $a = 1$. It supplies a monochromatic configuration of positive difference $d = m + n^2$ whose Y_0-coordinates lie in V_0. The first two torus conditions force

$$d\alpha = \gamma(\pi(T^d x)) - \gamma(\pi(x)) \in P_1 - P_0 \subseteq O_M.$$

Thus $d \in G_M$. The second case of Section 6 and the transfer back to the original coloring are unchanged. The geometric lemma is still applied only to the original nilsystem, so its dimension bound is not enlarged by joining the torus.

11.3 What simplifies in the unrestricted proof

Remark 11.3 (Unrestricted hierarchy versus localization). For the unrestricted hierarchy, Qiu removes the need to discuss the ideal of non-piecewise-syndetic sets in the recurrence argument. Every rescaled host is just $\mathbb{N}$, so a single previously constructed set $B = D_s$ can be reused at all packet stages instead of choosing fresh sets B_j. The geometric lemma needs only one candidate return time, not its syndeticity. These are genuine simplifications when only the strict color hierarchy is wanted.

For an arbitrary prescribed host, however, the conclusion $d = m+n^2$ does not ensure $d \in \mathcal{H}$. For example, the set $\{m + n^2 : n \in \mathbb{Z}\}$ itself is not piecewise syndetic and could be excluded by the host. The phase condition $d\alpha \in O_M$ does not correct that problem. Theorem 2.5 supplies the missing control in our main proof by allowing the prescribed excluded set to be removed from a piecewise syndetic supply of factor return times. There is also another way to obtain enough control from Qiu: retain the full set of linear progression differences and use its syndeticity, as we now explain.

11.4 Qiu plus syndeticity also gives localization

The following separate input is useful for an accurate comparison. It is the linear specialization of Lemma 3.6 in [3]; its proof there is independent of Theorem A.

Lemma 11.4 (Nonempty progression-return sets are syndetic). *Let (X, T) be minimal and invertible and let $U_0, \dots, U_{h-1} \subseteq X$ be nonempty and open. If*

$$R_h^X(U_0, \dots, U_{h-1}) \neq \varnothing,$$

then this return-time set is syndetic in $\mathbb{Z}$.

The cited lemma treats essentially distinct polynomial tuples, but only the linear case is needed here. To see the reduction behind it, choose one existing difference d_0 and put

$$V = \bigcap_{j=0}^{h-1} T^{-jd_0} U_j.$$

This is a nonempty open set. If $n \in R_h^X(V)$, then a point x with $T^{jn}x \in V$ satisfies $T^{j(n+d_0)}x \in U_j$ for every j. Thus

$$d_0 + R_h^X(V) \subseteq R_h^X(U_0, \dots, U_{h-1}).$$

The syndeticity input for the single-target set $R_h^X(V)$ gives the conclusion. The proof in [3, Lemma 3.6] obtains the required polynomial single-target input from the IP polynomial Szemerédi

theorem. We cite this input rather than treating syndeticity as a consequence of nonemptiness alone without justification.

Now use the torus-joined system and open targets U_j from the preceding Qiu argument, but allow an arbitrary host $\mathcal{H}$ again. The application of Qiu has already proved that

$$R := R_h^X(U_0, \dots, U_{h-1})$$

is nonempty. Lemma 11.4 makes R syndetic in $\mathbb{Z}$. For *every* $d \in R$, the defining first two targets force

$$d\alpha \in P_1 - P_0 \subseteq O_M.$$

Also every configuration witnessing membership in R is of the first color and lies above the desired nilsystem window. Consequently

$$R \cap \mathbb{N} \subseteq G_M$$

is syndetic in $\mathbb{N}$. Because $\mathbb{N} \setminus \mathcal{H}$ is not piecewise syndetic, Lemma 2.4 shows that

$$(R \cap \mathbb{N}) \cap \mathcal{H}$$

is piecewise syndetic and in particular nonempty. Choosing a difference there proves the localized first case. The second case, finite extraction, and the uniform induction with fresh sets B_j then proceed exactly as in the main proof.

Remark 11.5 (What this alternative does and does not establish). The full localized main theorem can therefore be proved using Qiu together with Lemma 11.4, without using Theorem A. The crucial set to which syndeticity is applied is the full linear-difference set R, not the particular differences of the form $m + n^2$ produced by the positivity device. There is no claim that those quadratic values form a syndetic set.

Theorem A still has two advantages for this exposition. It removes the torus joining and phase-target bookkeeping from the first case, and it gives the relative loss statement (6.4) for the original geometric candidate set P. The syndeticity shortcut only supplies a usable syndetic family somewhere in G_M; it does not compare all lifted return times to all factor return times. In particular, two arbitrary syndetic sets may be disjoint, so the shortcut by itself does not allow a prescribed piecewise syndetic subset of P to be retained. The general piecewise-syndetic boosting criterion in Proposition 10.1 is another place where the exceptional-set conclusion is directly useful.

11.5 The short modular separation

For the short modular separation in Section 10, Qiu also suffices without the general piecewise-syndetic criterion. When the first color projects onto all of Z, apply Lemma 11.1 with every $U_j = C_1$, with $m = 1$ and $a = 3$. The resulting difference $1 + 3n^2$ is positive and not divisible by three, hence belongs to S. In the other case, $\mathcal{P}_1$ of $3E$ supplies the monochromatic configuration inside the pure second-color cylinder exactly as before.

A Why the compact test list is sufficient

This appendix proves Lemma 7.1. Its purpose is to justify a countable list of compact parameter families, despite the uncountably many possible niltranslations. The essential observation is that the group and lattice models can be listed countably, while the translation parameter is handled by compactness within each model.

A.1 Move the starting point to the identity

Let $Y = G/\Gamma$, let T be left translation by $g_0 \in G$, and write the given starting point as $y = g\Gamma$. Left translation by g^{-1} sends y to the identity coset Γ and conjugates T to translation by

$$a = g^{-1} g_0 g.$$

Transport the open window by the same homeomorphism. Thus we can work with the identity coset as starting point without changing the form of the recurrence problem.

The identity component of G/Γ is

$$G^\circ/(G^\circ \cap \Gamma).$$

It has some finite period $m \geq 1$ under the translation by a, since the compact manifold has only finitely many components. Write

$$a^m = b\gamma, \qquad b \in G^\circ, \quad \gamma \in \Gamma.$$

On the identity component, the return map is

$$z\Gamma \longmapsto b\gamma z \gamma^{-1}\Gamma.$$

Pass to the connected simply connected covering group N of G° and let Λ be the inverse image of $G^\circ \cap \Gamma$. The resulting quotient N/Λ is the same connected nilmanifold. Conjugation by γ lifts to an automorphism A preserving Λ, and a lift of b gives the affine map

$$R_b(z\Lambda) = bA(z)\Lambda. \tag{A.1}$$

The starting point is still $o = e\Lambda$.

A.2 Why the automorphism is unipotent

Let

$$F(x) = \gamma x \gamma^{-1} x^{-1}$$

on the identity component. Repeated application of F produces iterated commutators with γ. Since G is nilpotent, some iterate is identically the identity element. The derivative at the identity satisfies

$$dF_e = \mathrm{Ad}(\gamma) - I.$$

Differentiating the constant iterate and using the chain rule gives

$$(\mathrm{Ad}(\gamma) - I)^c = 0$$

for some c. Thus the induced automorphism of the Lie algebra of N is unipotent. This is the meaning of unipotence for A in Lemma 7.1.

A.3 Why the affine map is still a niltranslation

Form the discrete semidirect product

$$G_A = N \rtimes_A \mathbb{Z}, \qquad \Gamma_A = \Lambda \rtimes_A \mathbb{Z}.$$

Its multiplication is

$$(x, r)(y, t) = (xA^r(y), r + t).$$

The quotient G_A/Γ_A identifies with N/Λ by choosing the representative $(x, 0)$. Under this identification, translation by $(b, 1)$ induces (A.1). The subgroup Γ_A is discrete and cocompact.

For completeness, we verify that G_A is nilpotent. Let $\mathfrak{n}$ be the Lie algebra of N, with lower central series $\mathfrak{n}_r$. The automorphism A preserves this series and is unipotent on each quotient $\mathfrak{n}_r/\mathfrak{n}_{r+1}$. On each such quotient choose a full flag lowered by $A - I$, as one can do from a Jordan basis for a unipotent linear map. Joining these flags, beginning with the deepest central quotient, gives

$$0 = F_0 \subset F_1 \subset \cdots \subset F_d = \mathfrak{n}$$

such that

$$[\mathfrak{n}, F_j] \subseteq F_{j-1}, \qquad (A - I)F_j \subseteq F_{j-1}.$$

The subspaces are A-invariant ideals. Their exponential subgroups satisfy

$$[G_A, \exp(F_j)] \subseteq \exp(F_{j-1}).$$

Indeed, conjugation by N is trivial on $\exp(F_j)/\exp(F_{j-1})$ by the bracket condition, and conjugation by the generator of $\mathbb{Z}$ is trivial there by the condition on $A - I$. Together with $G_A/N \cong \mathbb{Z}$, these subgroups give a central series for G_A.

The group $\mathbb{Z}$ is discrete, so $\dim G_A = \dim N$. Thus this presentation does not add a dimension to the nilmanifold. Every positive power R_b^e is translation by $(b, 1)^e$ on the same quotient.

A.4 Countably many models

Mal'cev's rational-coordinate theorem implies that a pair (N, Λ) can be described by a finite list of rational Lie structure constants and rational coordinates for a finite generating set of the lattice [6, Chapter II]. There are only countably many such finite rational descriptions. We may therefore choose a countable set of representatives for the isomorphism classes of pairs (N, Λ).

For a fixed pair, the lattice-preserving automorphisms form a countable set. They are determined by the images of finitely many lattice generators, each of which must lie in the countable group Λ. Uniqueness of the corresponding Lie-group automorphism follows because the logarithms of the lattice span the Lie algebra over $\mathbb{R}$ and N is simply connected. Retain the unipotent automorphisms. This gives countably many triples (N, Λ, A).

For each triple choose a compact exhaustion

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq N, \qquad \bigcup_{R \geq 1} K_R = N,$$

and a countable neighborhood basis $V_1, V_2, \ldots$ at $o = e\Lambda$. For instance, exponential images of larger and larger coordinate cubes give a compact exhaustion of N. Now list all tuples

$$(N, \Lambda, A, m, K_R, V_\ell), \qquad m, R, \ell \in \mathbb{N}.$$

This is the promised countable test list.

Given an original pointed nilsystem and window, perform the normalization and component reduction above. Choose R containing the resulting translation parameter b, and choose V_ℓ inside the transported window. The corresponding test represents the required recurrence problem at multiples of m. This proves Lemma 7.1.

A.5 Continuity used in finite extraction

For a fixed positive integer r, iterating (A.1) gives

$$R_b^r(z\Lambda) = bA(b)\cdots A^{r-1}(b)A^r(z)\Lambda.$$

The right side is continuous in b and z. Thus every fixed finite orbit-window condition is open in the parameter b. This supplies exactly the continuity required in Section 7; no uniform continuity in an unbounded iterate is asserted or needed.

References

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